

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |

Kent, OH - Open to relocation to Germany (visa sponsorship required)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) delivering enterprise AI agent systems end-to-end - LangChain/LlamaIndex-based RAG pipelines, multi-step agentic workflows, evaluation infrastructure, and production-grade containerized services. Master's degree (Kent State University). Measurable outcomes: ~35% retrieval precision improvement, >30% hallucination reduction, ~30% fewer post-deployment defects. Seeking Deployed Engineer role at LangChain to bridge the platform and DACH enterprise customers building production agent systems - bringing production deployment experience, not just prototype expertise.

CORE COMPETENCIES

LangChain / LangGraph • LlamaIndex • Retrieval-Augmented Generation (RAG) • Agentic Workflows • LLM Evaluation & Observability • Grounding & Guardrails • Python / FastAPI / Flask • Docker / CI-CD • Pre-Sales POC Design • Technical Stakeholder Communication • Production Deployment • Enterprise AI Delivery

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare AI, USA

- Developed **agentic LLM workflows** (LangChain) for multi-step healthcare queries (eligibility, care navigation, documentation clarification) - tool discipline, grounding rules, structured reasoning; reduced hallucination >**30%**, improved agent stability ~**25%**.
- Built **RAG systems** for clinical knowledge retrieval (LangChain + LlamaIndex): recursive semantic chunking, transformer-based embeddings, retrieval quality eval loop - precision improved ~**35%**; retrieval-grounded alignment exceeded **90%**.
- Built **LLM evaluation pipeline** with grounding rules, guardrails, hallucination tracking, and audit trails - reproducible quality assessment across production deployments.
- Shipped **FastAPI/Docker** production inference services with structured logging and load simulation - reduced post-deployment defects ~**30%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage, stakeholder system documentation.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Hyderabad, India

- Built **transformer-based video summarization system** - 60-70% reduction in manual review time; ~85% alignment with human-curated highlights.
- Implemented **document RAG Q&A** - reduced hallucinations ~30%, raised contextual accuracy >85%.
- Deployed as **Flask API with minimal UI** - non-technical staff operated independently without training.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India

- Built Jenkins CI/CD pipelines and Grafana monitoring dashboards - reduced deployment errors >30%, incident recurrence ~25%.
- Optimized T-SQL stored procedures for enterprise ERP workflows - ~20% query execution improvement.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate → Fraud Risk) with **schema-validated JSON contracts between agents** - LangGraph-equivalent orchestration pattern. RAG-grounded CPT/ICD validation, ANN similarity search for duplicate detection, explainable risk scoring with audit-ready traces.

LangChain, LlamaIndex, FastAPI, Docker, Python

Manga Lens - Chrome Extension for Real-Time AI Translation

shipped - Chrome Web Store

Full-stack browser extension integrating four AI providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-section capture pipeline, 7-day cache, per-domain configs for 29 sites. Built and shipped independently end-to-end.

TypeScript, React, Python, Manifest V3

Dream Decoder - Multimodal AI Platform

Full-stack FastAPI + React/TypeScript app with coordinated multimodal API integrations. Structured prompt transformation layers improved contextual alignment ~30%.

FastAPI, React, TypeScript, multi-modal AI APIs

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL (PostgreSQL, T-SQL, SQLite)

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, YOLOv8

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs

Work Authorization: F-1 OPT (US-authorized); requires visa sponsorship for Germany/EU